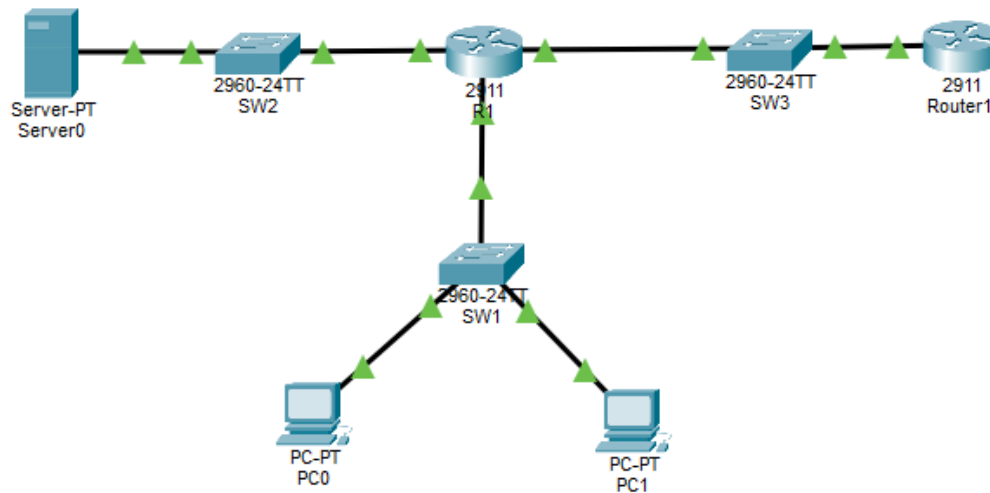


TOPOLOGY



97.

```
Router>ena
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int g0/0
Router(config-if)#ip address 10.10.20.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state
up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 10.10.20.1 10.10.20.10
Router(config)#ip dhcp pool EXTERNAL-LAN
Router(dhcp-config)#network 10.10.20.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.20.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.20.2      YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
```

98.

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 10.10.10.1 10.10.10.10
Router(config)#ip dhcp pool INTERNAL-LAN
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#default-router 10.10.10.1
Router(dhcp-config)#dns-server 10.10.20.10
Router(dhcp-config)#end
```

PC1 IP ADDRESS

IPv4 Address	10.10.10.11
Subnet Mask	255.255.255.0

PC2 IP ADDRESS

IPv4 Address	10.10.10.12
Subnet Mask	255.255.255.0

99.

External server DHCP settings

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: EXTERNAL-LAN

Default Gateway: 10.10.10.1

DNS Server: 10.10.20.10

Start IP Address: 10 10 10 11

Subnet Mask: 255 255 255 0

Maximum Number of Users: 245

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
EXTERNAL-LAN	10.10.10.1	10.10.20.10	10.10.10.11	255.255.255.0	245	0.0.0.0	0.0.0.0

R1 Helper address

```
Router#ena
Router#conf t
Enter configuration commands, one per line. End w
Router(config)#int g0/0
Router(config-if)#ip helper-address 10.10.20.10
Router(config-if)#no shutdown
Router(config-if)#end
_
```

PC1

IP Configuration	
☒ DHCP	☐ Static
DHCP failed. APIPA is being used.	
IPv4 Address	169.254.202.134
Subnet Mask	255.255.0.0

PC2

IP Configuration	
☒ DHCP	☐ Static
DHCP failed. APIPA is being used.	
IPv4 Address	169.254.176.186
Subnet Mask	255.255.0.0

Reflection Question

100. An Ip helper-address is needed when the DHCP server is on a different subnet because DHCP requests are broadcasts, and routers do not normally forward broadcasts between networks. The helper address forwards these requests to the DHCP server. If the server is on the same subnet as the client, the broadcast can reach it directly, so a helper address is not needed.